

Sparse inelastic contact kernels: model and numerical contract

Physics-engine implementation note

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The implementation is planar and acts on an already discovered, frozen contact graph. The accompanying measured report establishes kernel performance, not complete-engine throughput or a new theory of collision. Geometry, masses, inertias and material coefficients remain unchanged when selecting an algorithm.

1 Body and contact operators

For each body use $\tilde{U}_i = (v_{ix}, v_{iy}, \ell\omega_i)$, where the arbitrary reference length $\ell > 0$ is a representation choice. Inverse mass is stored as a vector rather than a dense diagonal matrix:

$$H_i^{-1} = \text{diag}(m_i^{-1}, m_i^{-1}, \ell^2/\mathbb{I}_i). \quad r^\wedge = \begin{bmatrix} -r_y & r_x \end{bmatrix}. \quad (1)$$

At an edge going from B to A, a direction d contributes the row

$$G_d = \begin{bmatrix} d_x & d_y & (r_A^\wedge d)/\ell & -d_x & -d_y & -(r_B^\wedge d)/\ell \end{bmatrix}. \quad (2)$$

Use one normal and tangent row per point, not one summed impulse per body pair. Prescribed degrees of freedom have zero inverse mass and retain their supplied velocity in the relative-motion map. Sparse operators give

$$V^- = G\tilde{U}^-, \quad K = GH^{-1}G^\top, \quad V^+ = V^- + K\Delta P, \quad \tilde{U}^+ = \tilde{U}^- + H^{-1}G^\top\Delta P. \quad (3)$$

No inverse of the full, potentially singular contact mobility is required. World geometry/mass changes invalidate this fixed-geometry cache. Sparse factorisation may have substantial fill on highly connected graphs; linear row scaling is not a theorem about all contact graphs.

2 Normal active-set solve

For zero restitution and frictionless normals, solve the standard convex QP

$$\min_{p_n \geq 0} \frac{1}{2} p_n^\top K_{nn} p_n + (V_n^-)^\top p_n. \quad V_n^+ \geq 0, \quad p_n \odot V_n^+ = 0. \quad (4)$$

Free active rows are solved together; a candidate crossing a bound moves only to its first bound. Wrong bound gradients release the corresponding variables. Singular free blocks use least squares without introducing compliance. Cycling, inconsistent stationarity or excessive work rejects the candidate. This block scheme has no claimed universal convergence guarantee.

The acceptance gate checks nonnegative impulses, normal feasibility and the velocity at contacts carrying positive impulse. Raw complementarity products are also reported; their scale grows with impulse magnitude. The benchmark requires every ball velocity within 10^{-8} m/s and wall reaction within 10^{-5} kg m/s. Small islands may choose dense factorisation of the same QP; the contact-count threshold is a numerical policy, not a physical parameter.

3 Implicit Coulomb residual and exact structural dispatch

One nonnegative coefficient μ defines the implemented inelastic law:

$$p_n \geq 0, \quad V_n^+ \geq 0, \quad p_n V_n^+ = 0, \quad (5)$$

$$|p_t| \leq \mu p_n, \quad V_t^+ \neq 0 \implies p_t = -\mu p_n \text{sign}(V_t^+). \quad (6)$$

At $V_t^+ = 0$, friction may lie inside its capacity. Final sticking does not constitute elastic tangential rebound. No separate static/dynamic coefficients, elastic-history state, positive restitution or free rolling couple is provided.

Positive diagonal numerical scales ρ_n, ρ_t give an equivalent root system:

$$F_n(p) = p_n - \max(0, p_n - \rho_n V_n^+), \quad (7)$$

$$F_t(p) = p_t - \text{clip}(p_t - \rho_t V_t^+, -\mu \max(p_n, 0), \mu \max(p_n, 0)). \quad (8)$$

The implementation uses a sparse generalised Jacobian and a merit line search. A disclosed sparse least-squares fallback is allowed; optimiser success is never a replacement for mechanical residual checks. Four held-out stress cases remain rejected. No existence or uniqueness conclusion follows from rejection.

If and only if the assembled cross-block has exact structural zero, $K_{nt} = 0$, the normal pressure can be solved first and tangential response is the bounded QP

$$\min_{-\mu p_n \leq p_t \leq \mu p_n} \frac{1}{2} p_t^\top K_{tt} p_t + (V_t^-)^\top p_t. \quad (9)$$

Its KKT conditions give sticking or saturated friction opposite final slip. This is the same implicit law in this decoupled case, not a conic-QP replacement for general hard Coulomb contact. Small nonzero cross terms are not dropped. Nonunique normal pressures can produce different friction outcomes; the selected active-set pressure distribution is not an authenticated material force history.

4 Mechanical passivity with driven walls

Separate prescribed relative motion as $V_{\text{prescribed}}$. The instantaneous external work and change in dynamic-body kinetic energy obey

$$W_{\text{boundary}} = -V_{\text{prescribed}}^\top \Delta P, \quad (10)$$

$$\Delta T - W_{\text{boundary}} = (V^-)^\top \Delta P + \text{frac}12 \Delta P^\top K \Delta P \quad (11)$$

$$= (V^+)^\top \Delta P - \text{frac}12 \Delta P^\top K \Delta P \leq 0. \quad (12)$$

The last inequality follows from normal complementarity, dissipative Coulomb slip and $K \succeq 0$. The implementation also checks it numerically. Content kinetic energy may increase when the actuator supplies work.

For the closed unit-mass row driven at 1 m/s, $1 \leq v_{1x}^+ \leq \dots \leq v_{Nx}^+ \leq 1$ forces every ball to 1 m/s. Thus momentum is N , kinetic energy $N/2$, wall work N and dissipation $N/2$ in the corresponding SI units. The independent audit uses explicit row force, torque and point-velocity formulas rather than the new sparse assembler.

5 Scope of the result

The performance report retains every timing sample, 139 snapshots, source archives and rejected inputs. Long-row gains justify investigating a sparse island kernel in the actual engine. They do not establish full moving-container convergence, accurate stress/elastic waves, continuous collision discovery or a universal solver. The coefficients are synthetic. The compiler handover retains language unit/shape semantics and independent native/WASM/hardware acceptance boundaries.

Established prior art includes Alart–Curnier (1991), [https://doi.org/10.1016/0045-7825\(91\)90022-x](https://doi.org/10.1016/0045-7825(91)90022-x), and Anitescu–Potra (1997), <https://doi.org/10.1023/a:1008292328909>. Bibliographic metadata was checked; the particular implementation is independently verified through its declared mechanical residuals and tests, not assumed identical to either paper.